

Integers and Absolute Value

Lesson 9-2

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*... -3, -2, -1, 0, 1, 2, 3 ... shown on a number line
with 0 in the center*

Integer

A whole number that can be positive, negative, or zero.

*+5 means 5 above zero, like 5 floors above ground
level*

Positive

A number bigger than zero, to the right of zero on a number line.

-3 means 3 below zero, like 3 floors underground

Negative

A number smaller than zero, to the left of zero on a number line.

$|-5| = 5$ and $|5| = 5$ — both are 5 units from 0

Absolute value

How far a number is from zero. It is never negative.

3 and -3 are opposites — both are 3 units from 0

Opposite

Two numbers the same distance from zero, on opposite sides, like 3 and -3.

Key Ideas & Notes

- Captain Vega's treasure map has clues hidden at different elevations!
- Some treasures are buried underground (below sea level) and some are on mountaintops (above sea level).
- Sea level is marked as 0.
- A treasure chest is at -4 (4 meters below sea level) and a lookout tower is at +6 (6 meters above sea level).
- Plot each integer on the number line and find its absolute value (distance from zero).

Think About It

- What does the negative sign tell you about the treasure chest's location?
- Which location is farther from sea level — the chest at -4 or the tower at +6?
- How can you find the distance from sea level without worrying about direction?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Why are -5 and 5 both the same distance from zero on the number line, even though one is negative and one is positive?

Level 1 support

Start here: -5 and 5 are the same distance from zero because ____.

Both numbers are ____ units from zero.

Ambos numeros estan a ____ unidades del cero.

They are different because one is ____ and one is ____.

Son diferentes porque uno es ____ y el otro es ____.

Word bank: integer absolute value opposite positive negative

Listen for: Student says both are 5 units from zero (same absolute value) but are opposites because of their signs.

Level 2

Push students to explain why absolute value can never be negative.

- Absolute value is never negative because ____.
- The absolute value of any number tells me ____.

EXPLORE

A diver is at -30 feet and a hawk is at +30 feet. How can absolute value help you describe how far each is from sea level?

Level 1 support

Start here: Absolute value tells me how far each animal is from ____.

The diver is ____ feet from sea level because $|-30| =$ ____.

El buzo esta a ____ pies del nivel del mar porque $|-30| =$ ____.

The hawk is ____ feet from sea level because $|30| =$ ____.

El halcon esta a ____ pies del nivel del mar porque $|30| =$ ____.

Word bank: (integer) (absolute value) (positive) (negative) (opposite)

Listen for: Student uses absolute value to show both are 30 feet from sea level (zero), ignoring direction.

Level 2

Ask students to decide who is truly farther from sea level and justify using both the integers and their absolute values.

- They are the same distance from sea level because ____.
- The sign tells me the direction, while the absolute value tells me ____.

CONNECT

How is the opposite of a number different from the absolute value of a number?

Level 1 support

Start here: The opposite changes ____, but the absolute value gives ____.

The opposite of 7 is ____.

El opuesto de 7 es ____.

The absolute value of -7 is ____.

El valor absoluto de -7 es ____.

Word bank: integer absolute value opposite positive negative

Listen for: Student explains the opposite flips the sign while absolute value always gives the non-negative distance from zero.

Level 2

Push students to find a number whose opposite and absolute value are equal, and explain why.

- The opposite and absolute value are equal when the number is ____.
- This works because ____.

REFLECT

Think of a real situation where a negative integer makes sense. What does zero mean in that situation?

Level 1 support

Start here: A negative integer makes sense when ____, and zero means ____.

A negative number could describe ____.

Un numero negativo podria describir ____.

In that situation, zero means ____.

En esa situacion, el cero significa ____.

Word bank: integer negative positive absolute value opposite

Listen for: Student gives a real context (temperature, elevation, money) and explains zero as the reference point between positive and negative.

Level 2

Ask students to critique whether absolute value would be useful in their chosen situation and why.

- Absolute value would help in my situation because ____.
- It would tell me ____ no matter the direction.

Guided Examples

Example 1

What is the absolute value of -9?

Solution: Absolute value is the distance from zero. -9 is 9 units from zero, so $|-9| = 9$.

Answer: A. 9

Example 2

Which integer has the GREATEST absolute value: -3, 7, -5, 2?

Solution: $|-3| = 3$, $|7| = 7$, $|-5| = 5$, $|2| = 2$. The greatest absolute value is 7.

Answer: A. 7

Example 3

What is $|0|$?

Solution: Zero is 0 units from itself on the number line, so $|0| = 0$.

Answer: A. 0

Write About the Math

The Writing Revolution

I can explain my reasoning using the words integer, positive, negative, absolute value, and opposite.

1. Kernel Sentence subject + verb

Model: Integer is a whole number that can be positive, negative, or zero.

Número entero es un número entero que puede ser positivo, negativo o cero.

Write a kernel sentence about integer. Use a subject and a verb.

Escribe una oración base sobre número entero. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Integer matters in math

Número entero importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Integer matters in math because ____.

Número entero importa en matemáticas porque ____.

but
pero

Integer matters in math, but ____.

Número entero importa en matemáticas, pero ____.

so
entonces

Integer matters in math, so ____.

Número entero importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about integer.

Di un hecho verdadero sobre integer.

Integer ____.

Question *Pregunta*

Ask a question about integer.

Haz una pregunta sobre integer.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about integer.

Muestra entusiasmo sobre integer.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with integer.

Dile a un compañero qué hacer con integer.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

Both numbers are ____ units from zero.

Ambos numeros estan a ____ unidades del cero.

They are different because one is ____ and one is ____.

Son diferentes porque uno es ____ y el otro es ____.

The diver is ____ feet from sea level because $|-30| =$ ____.

El buzo esta a ____ pies del nivel del mar porque $|-30| =$ ____.

| Try It

Solve on your own. Check the answer key when you are done.

1. What is $|0|$?

- A. 0
- B. 1
- C. -1
- D. There is no answer

Show your work:

2. Which statement about absolute value is TRUE?

- A. $|-4| = |4|$ because both are 4 units from zero
- B. $|-4| = -4$ because the number is negative
- C. $|4| > |-4|$ because positive is always bigger
- D. $|-4| = 0$ because negatives cancel out

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A submarine is at -150 meters and a bird is flying at +150 meters. A student says the submarine is closer to sea level because -150 is less than 150. Is the student correct? Explain using absolute value.

Sentence starter: The student is ____ because absolute value measures _____. Since $|-150| = \underline{\hspace{1cm}}$ and $|150| = \underline{\hspace{1cm}}$, both are ____ from sea level. The student confused ____ with ____.

Show your work:

Reflect — Exit Ticket

Which integer has an absolute value of 8 and is located to the LEFT of zero on the number line?

- A. 8
- B. -8
- C. 0
- D. -80

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 0 — *Zero is 0 units from itself on the number line, so $|0| = 0$.*
2. **Try It 2:** A. $|-4| = |4|$ because both are 4 units from zero — *Absolute value measures distance from zero. Both -4 and 4 are 4 units from zero, so $|-4| = |4| = 4$.*
3. **Exit Ticket:** B. -8 — *To the left of zero means negative. The integer -8 has an absolute value of 8 because it is 8 units from zero.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Integer is a whole number that can be positive, negative, or zero.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).